Electrical Actuators and Drives

***Abstract*—This report provides a systematic technical investigation of primary electromechanical actuators and electric motor configurations. The analysis evaluates physical construction, operational principles, torque production mechanisms, topological variants, advantages, disadvantages, and real-world industrial deployments. Actuators covered include Brushed DC, Brushless DC (BLDC), Permanent Magnet Synchronous Motors (PMSM), Stepper Motors, Conventional Servomechanisms, Smart Bus Actuators, AC Induction Machines, Synchronous Motors, Reluctance Motors (SRM/SynRM), and Universal Motors.**

***Keywords*—Actuators, Back-EMF, BLDC, Induction Motors, Lorentz Force, PMSM, Reluctance Motors, Servos, Smart Actuators, Stepper Motors.**

I. INTRODUCTION

Electric motors convert electrical energy into mechanical rotation or translation through electromagnetic field interactions established between a stationary stator and a rotating rotor. These systems operate fundamentally on the Lorentz Force Law, Faraday's Law of Electromagnetic Induction, and the physical tendency of magnetic circuits toward configurations of minimum reluctance. Modern actuation systems range from basic mechanical commutation units to micro-controlled digital vector drives.

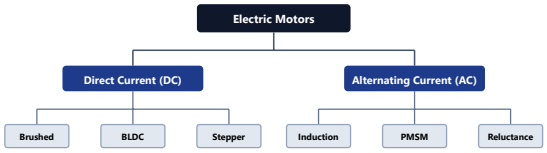


Fig. 1. Categorisation of Electric Motors

II. BRUSHED DC MOTORS (BDC)

A. Construction & Components

A brushed DC machine consists of a stationary field frame (stator) housing permanent magnets or wound electromagnets, and an internal rotating armature (rotor) composed of a slotted, laminated silicon-steel core wound with insulated copper magnet wire. Current transfer is handled by mechanical carbon or graphite brushes mounted in spring-loaded brush holders, maintaining continuous sliding contact against a segmented copper commutator mounted on the central drive shaft.

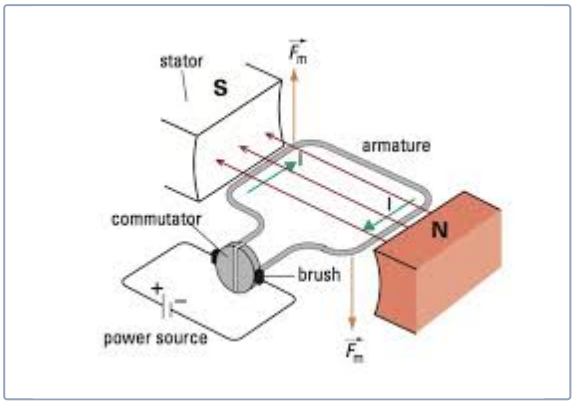


Fig. 2. Workinf od a simple DC Motor.

B. Working Principle & Torque Production

Direct current is supplied through external terminals to the stationary brushes, flowing directly into the commutator bars and through active armature coils. Immersed within the stator magnetic flux Φ, the current-carrying conductors experience a localized Lorentz force ($d\mathbf{F} = I_a(d\mathbf{l} \times \mathbf{B})$). Because current flows in opposite directions on each side of a coil loop, opposing tangential forces produce net shaft torque T. As the shaft rotates through the magnetic neutral plane, the commutator segments physically swap brush contact, reversing the current direction within the coil to maintain continuous unidirectional rotation.

The gross shaft output torque T and induced counter-electromotive force (Back-EMF) E_b are governed by:

$$T = k_t \Phi I_a, \quad E_b = k_e \Phi \omega_m \tag{1}$$

where k_t is the motor torque constant, k_e is the Back-EMF constant, I_a is armature current, and ω_m is rotor angular speed. The terminal voltage balance is given by $V_t = E_b + I_a R_a$.

C. Types & Variations

- **Permanent Magnet (PMDC):** Field produced by permanent magnets; linear torque-speed curve; compact form-factor.
- **Series-Wound:** Field coil in series with armature; high starting torque ($T \propto I_a^2$); runaway speed under zero load.
- **Shunt-Wound:** Field wired in parallel; near-constant speed regulation over shifting loads.
- **Compound-Wound:** Combines series and shunt field windings to prevent runaway while retaining robust breakaway torque.

D. Advantages & Disadvantages

- *Advantages:* Simple linear voltage control; low drive electronics complexity; high starting torque;

low manufacturing cost.

- *Disadvantages:* Mechanical brush wear requiring routine maintenance; sparking and arcing; severe electromagnetic interference (EMI); limited maximum speed.

E. Applications & Real-World Examples

Automotive starter motors, power window lifts, windshield wiper drives, portable power tools, low-cost battery toys, and mobility wheelchairs.

III. BRUSHLESS DC MOTORS (BLDC)

A. Construction & Components

BLDC motors invert the brushed arrangement: high-energy permanent magnets (such as NdFeB or SmCo) are mounted directly onto the rotating rotor, while three-phase insulated stator coils reside in a stationary slotted iron frame. Commutation relies entirely on solid-state electronics, typically utilizing three digital Hall-effect sensors spaced 120° electrical apart or sensorless back-EMF zero-crossing networks.

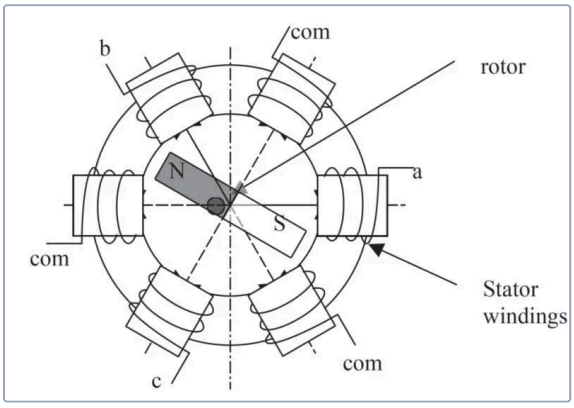


Fig. 3. BLDC diagram.

B. Working Principle & Torque Production

Electronic commutation utilizes a 3-phase full-bridge inverter (six MOSFETs/IGBTs). Based on instantaneous rotor position feedback, pairs of stator phases are energized in a 6-step trapezoidal sequence (120° electrical conduction). This produces an advancing stator magnetic field vector that remains approximately 90° ahead of the rotor magnets. Torque is generated through magnetic attraction and repulsion pulling the rotor magnets toward the energized stator poles.

C. Types & Variations

- **Inrunner:** Rotor spins inside the stator core; low rotor inertia enables rapid angular acceleration.
- **Outrunner:** Stator windings are stationary in the center, and the outer magnet-lined bell rotates; delivers higher torque without an inline gearbox.
- **Slotless/Coreless:** Stator lacks iron teeth; eliminates cogging torque for ultra-smooth rotation.

D. Advantages & Disadvantages

- *Advantages:* High efficiency (85–92%); no brush wear or sparking; high power-to-weight ratio; effective heat dissipation through the external stator housing.
- *Disadvantages:* Requires a multi-transistor electronic speed controller (ESC); higher manufacturing cost; trapezoidal commutation torque ripple.

E. Applications & Real-World Examples

Multirotor drone propellers (DJI UAVs), electric bicycles, computer hard drive spindles, cordless power drills, and medical centrifuges.

IV. PERMANENT MAGNET SYNCHRONOUS MOTORS (PMSM)

A. Construction & Components

A PMSM features sinusoidally distributed three-phase stator windings within slotted steel laminations and high-energy permanent magnets embedded or mounted on the rotor. Dynamic vector control requires high-resolution position feedback from optical encoders, magnetic absolute encoders, or resolvers.

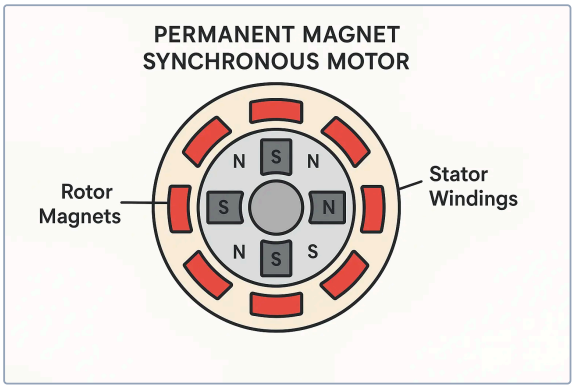


Fig. 4. PMSM diagram.

B. Working Principle & Torque Production

Balanced three-phase sinusoidal AC currents energize the stator, creating a smooth, continuously revolving magnetic field. Driven by Field-Oriented Control (FOC), stator phase currents are transformed into orthogonal direct (i_d, flux-producing) and quadrature (i_q, torque-producing) vector components. Torque is produced via two mechanisms: magnetic alignment torque (interaction of permanent magnet flux with i_q) and reluctance torque arising from differences between direct and quadrature inductances ($L_q - L_d$) in salient rotors.

C. Types & Variations

- **Surface Permanent Magnet (SPM):** Magnets mounted on rotor surface; non-salient ($L_d \approx L_q$); produces pure magnetic alignment torque.
- **Interior Permanent Magnet (IPM):** Magnets embedded inside rotor laminations; exhibits magnetic saliency ($L_q > L_d$), delivering reluctance torque and enabling high-speed flux-weakening.

D. Advantages & Disadvantages

- *Advantages:* Ultra-high efficiency (92–97%); smooth torque output with near-zero ripple; high

power density; wide operational speed envelope via field weakening.

- *Disadvantages:* Requires high-speed DSP vector control; higher rare-earth magnet costs; risk of permanent thermal demagnetization under excessive temperatures.

E. Applications & Real-World Examples

Electric vehicle traction drives (Tesla Model 3/Y, Toyota Prius MG2), industrial articulated robotics, CNC machine tool spindles, and high-speed rail traction.

V. STEPPER MOTORS

A. Construction & Components

A stepper motor features a multi-phase stator with salient electromagnetic poles and a finely toothed iron rotor (incorporating an axially magnetized permanent magnet disc in hybrid configurations). It operates without mechanical brushes, using dual H-bridge switching drivers to sequence current through individual phase coils.

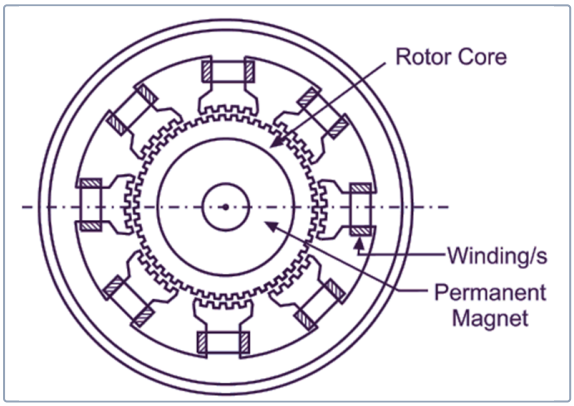


Fig. 5. Cutaway of a hybrid stepper motor revealing toothed magnetic rotor cups and multi-phase stator poles.

B. Working Principle & Torque Production

Operating as an open-loop digital actuator, sequential electrical pulses energize stator phase coils in sequence. Rotor teeth are pulled toward the energized stator teeth by magnetic reluctance forces seeking the path of minimum magnetic gap resistance. When stator coils remain energized while stationary, the motor produces holding torque, locking the rotor in position.

The full step angle θ_s is governed by phase count m and rotor teeth count Z_r:

$$\theta_s = \frac{360^\circ}{m \cdot Z_r} \tag{2}$$

A standard 2-phase hybrid stepper with $Z_r = 50$ yields a fundamental 1.8° step angle. Microstepping modulates adjacent phase currents sinusoidally to subdivide steps and suppress low-speed resonance.

C. Types & Variations

- **Variable Reluctance (VR):** Unmagnetized soft-iron toothed rotor; rotation driven purely by reluctance alignment.
- **Permanent Magnet (PM):** Cylindrical magnetized rotor; larger step angles (7.5°–15°).
- **Hybrid Synchronous:** Combines VR toothed rotor cups with an internal permanent magnet core, producing precise 1.8° and 0.9° steps.

D. Advantages & Disadvantages

- *Advantages:* Deterministic open-loop positioning without encoder feedback; full holding torque at zero RPM; direct digital pulse control; long operational lifespan.
- *Disadvantages:* Torque drops off sharply at higher speeds; step loss occurs if load exceeds pull-out torque; prone to low-speed resonance; draws continuous holding current.

E. Applications & Real-World Examples

Desktop 3D printers (X/Y/Z motion axes and filament extruders), CNC routers, flatbed document scanners, camera pan-tilt heads, and medical fluid infusion pumps.

VI. CONVENTIONAL SERVOMOTORS

A. Construction & Components

A conventional servomotor integrates a core DC or miniature brushless drive motor, a high-reduction gear train (metal or nylon), a potentiometer sensor mechanically linked to the output horn, and an internal analog or digital closed-loop error amplifier board.

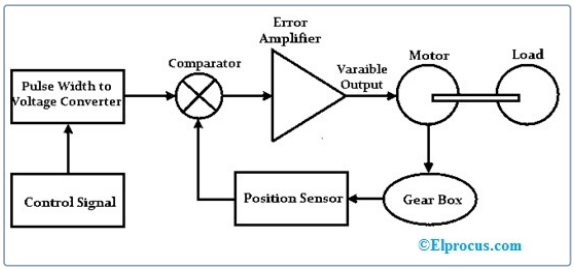


Fig. 6. Standard servomechanism: core DC drive motor, gear cluster, and potentiometer feedback.

B. Working Principle & Torque Production

Target positions are commanded through a standard 50 Hz PWM signal (pulse widths typically 1.0 ms to 2.0 ms). The onboard controller evaluates the error between the commanded pulse width and the potentiometer's feedback voltage. An internal H-bridge powers the core motor with a drive voltage proportional to the positional error, driving the gear reduction train until the error reaches zero. Output torque equals motor torque multiplied by gear reduction ratio and mechanical efficiency.

C. Types & Variations

- **Analog Servos:** 50 Hz refresh rate; analog comparator driver.
- **Digital Servos:** High-frequency microcontroller drive (300+ Hz); faster response and stiffer holding deadband.
- **Continuous Rotation:** Internal stop pin and feedback loop modified to provide variable speed

instead of absolute position locking.

D. Advantages & Disadvantages

- *Advantages:* Integrated closed-loop position control; compact packaging; simple 3-wire interface (Power, Ground, PWM); high holding torque.
- *Disadvantages:* Mechanical gear backlash; limited angular travel (typically 180°–270°); potentiometer wiper wear; unidirectional communication (no operational telemetry returned).

E. Applications & Real-World Examples

Radio-controlled aircraft control surfaces (ailerons/elevators), RC vehicle steering linkages, hobby robotic arms, and automated camera mounts.

VII. SMART SERVOS & BUS ACTUATORS

A. Construction & Components

Smart servos integrate a high-performance coreless or brushless DC motor, steel/titanium gear set, 12-to-16-bit absolute magnetic encoder, onboard 32-bit ARM Cortex processor, full-bridge driver, and a multi-drop half-duplex serial bus interface (TTL UART, RS-485, or CAN FD).

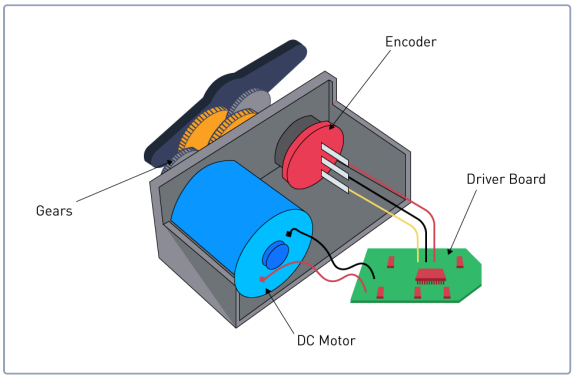


Fig. 7. Servo Motor from inside.

B. Working Principle & Torque Production

Actuators are daisy-chained on a shared serial bus with individual network IDs. The host sends addressed command packets instructing position, speed, acceleration profiles, or torque limits. The internal MCU runs localized PID loops, driving the motor while streaming back real-time operating metrics.

Torque is directly regulated by closed-loop current feedback. By monitoring winding currents, the processor adjusts torque dynamically, enabling programmable compliance, virtual spring-damper characteristics, and electronic collision detection.

C. Types & Variations

- **Modular Robotic Bus Servos:** Dynamixel (AX/MX/X-series), Feetech STS; targeted at multi-joint articulated robotics.
- **Integrated Industrial Actuators:** High-torque brushless motors paired with zero-backlash cycloidal/harmonic drive gearing and CANopen/EtherCAT bus control.

D. Advantages & Disadvantages

- *Advantages:* Full bidirectional telemetry reporting position, speed, current, temperature, and diagnostic errors; daisy-chain wiring significantly reduces harness bulk; programmable compliance; continuous 360° multi-turn operation with absolute encoding.
- *Disadvantages:* Higher per-unit cost; serial bus polling latency on large networks; software protocol integration overhead.

E. Applications & Real-World Examples

Humanoid bipedal robots (DARwIn-OP, Robotis OP3), quadruped robot limbs, multi-finger prosthetic hands, and industrial pick-and-place arms.

VIII. AC INDUCTION MOTORS

A. Construction & Components

An AC induction motor features a stationary laminated steel stator with three-phase distributed windings and a rugged rotor core. The dominant rotor design is the "Squirrel-Cage," constructed with heavy conductive aluminum or copper bars shorted at both axial ends by robust continuous end-rings.

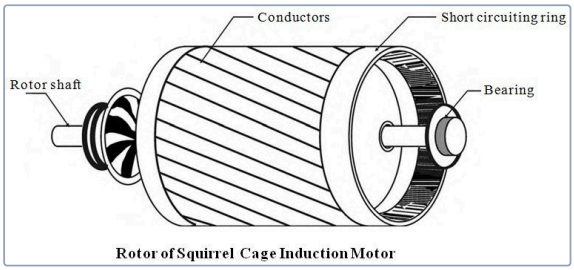


Fig. 8. Squirrel-cage conductive rotor structure displaying conductive bars shorted by robust end rings.

B. Working Principle & Torque Production

Three-phase AC stator currents produce a magnetic field rotating at synchronous speed N_s:

$$N_s = \frac{120f}{P} \tag{3}$$

where f is electrical frequency and P is stator pole count. This revolving field cuts the rotor conductors, inducing voltages by Faraday's Law. The resulting induced currents generate rotor magnetic poles that interact with the stator field via Lenz's Law, producing rotational torque. To maintain continuous flux cutting, the rotor must rotate slower than synchronous speed; this relative speed difference is termed slip s:

$$s = \frac{N_s - N_r}{N_s} \tag{4}$$

Torque output is zero at synchronous speed ($s = 0$) and increases approximately linearly with slip up to the motor's breakdown torque limit.

C. Types & Variations

- **Squirrel-Cage (SCIM):** Standard, highly durable rotor design; widely used across industrial drives.
- **Wound-Rotor (WRIM / Slip Ring):** Rotor windings accessible via slip rings; external

resistance can be added to provide high breakaway starting torque with reduced inrush current.

- **Single-Phase Induction:** Requires starting capacitor or shaded-pole configuration to establish an initial revolving field.

D. Advantages & Disadvantages

- *Advantages:* Highly durable and low maintenance; lowest cost per kilowatt for continuous duty; self-starting on utility 3-phase power; operates safely in harsh environments.
- *Disadvantages:* Inherent slip losses reduce efficiency; low lagging power factor under light loads; high starting current inrush (500–700% of rated current); requires a Variable Frequency Drive (VFD) for variable speed control.

E. Applications & Real-World Examples

Industrial water pumps, HVAC blowers and chillers, centrifugal gas compressors, municipal wastewater treatment plants, and mining conveyor systems.

IX. SYNCHRONOUS MOTORS (EXCITED-ROTOR)

A. Construction & Components

A synchronous motor features a three-phase distributed AC stator and an electromagnetic rotor energized with direct current (DC) via slip rings and brushes or an integrated brushless exciter. Damper (amortisseur) windings are embedded in the rotor pole faces to enable initial line starting and suppress mechanical oscillations.

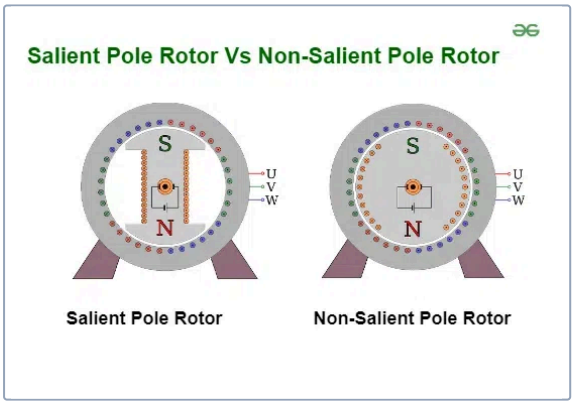


Fig. 9. Salient-pole synchronous rotor vs Non- Salient-pole synchronous rotor

B. Working Principle & Torque Production

The three-phase stator produces a rotating magnetic field revolving at synchronous speed $N_s = 120f/P$. Direct current energizes the rotor field coils, producing fixed magnetic north and south poles. These rotor poles lock into magnetic synchronism with the rotating stator field, forcing the rotor to turn at exact synchronous speed with zero steady-state slip ($N_r = N_s$). Shaft torque is proportional to the magnetic field strength and the sine of the torque angle δ separating the rotor and stator field axes.

C. Types & Variations

- **Salient-Pole:** Projecting poles; lower speed; suitable for high pole count industrial machinery.
- **Cylindrical Rotor:** Smooth round rotor; high-speed 2-pole or 4-pole machinery.
- **Synchronous Condenser:** Unloaded synchronous motor used purely to provide leading reactive power (VAr) for plant power factor correction.

D. Advantages & Disadvantages

- *Advantages:* Constant rotational speed regardless of varying shaft load (up to pull-out torque); adjustable power factor (can run leading to correct factory power factor); high efficiency in megawatt-scale applications.
- *Disadvantages:* Not inherently self-starting (requires damper winding run-up or VFD starting); requires separate DC excitation source; susceptible to hunting or pull-out under sudden load transients.

E. Applications & Real-World Examples

Industrial ball mills in cement processing, large reciprocating gas compressors, metal rolling mills, electrical utility synchronous condensers, and marine propulsion drives.

X. RELUCTANCE MOTORS (SRM & SYNRM)

A. Construction & Components

Reluctance machines use simple soft-magnetic steel laminations in the rotor, completely eliminating permanent magnets, rotor copper windings, and squirrel-cage bars. Switched Reluctance Motors (SRM) feature doubly-salient stator and rotor poles, while Synchronous Reluctance Motors (SynRM) employ distributed AC stators with rotor flux barriers stamped to direct magnetic flux lines.

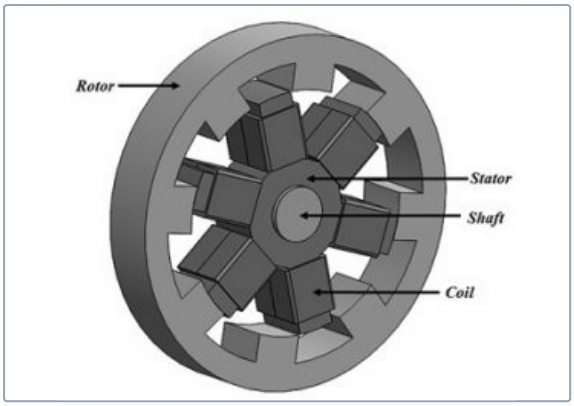


Fig. 10. Synchronous reluctance rotor.

B. Working Principle & Torque Production

The rotor naturally rotates toward angular positions that minimize magnetic reluctance (maximizing magnetic permeance and phase inductance). In SRMs, phases are energized sequentially as rotor teeth approach alignment. In SynRMs, a conventional rotating AC field pulls the rotor flux barrier profile smoothly around at synchronous speed. Torque is produced by the gradient of phase inductance relative to rotor angle θ:

$$T = \frac{1}{2} i^2 \frac{dL(\theta)}{d\theta} \tag{5}$$

Torque magnitude depends on phase current and the rate of change of inductance, operating independently of current polarity.

C. Types & Variations

- **Switched Reluctance Motor (SRM):** Doubly-salient poles; driven by dedicated asymmetric half-bridge converters.
- **Synchronous Reluctance Motor (SynRM):** Distributed stator; driven by standard sinusoidal FOC inverters.
- **Permanent Magnet-Assisted SynRM:** Ferrite magnets inserted in flux barriers to boost power factor without rare-earth materials.

D. Advantages & Disadvantages

- *Advantages:* 100% free of rare-earth permanent magnets; zero rotor copper losses; mechanically rugged rotor capable of extreme rotational speeds and high temperatures; low rotor thermal footprint.
- *Disadvantages:* Higher acoustic noise and vibration in SRMs; non-linear torque ripple; lower natural power factor compared to PMSM, requiring slightly larger inverters.

E. Applications & Real-World Examples

High-efficiency industrial pump drives (ABB SynRM IE5 systems), ultra-high-speed vacuum impellers (Dyson digital motors), severe-environment mining equipment, and modern rare-earth-free electric vehicle drivetrains.

XI. SPECIAL ACTUATOR TOPOLOGIES

A. Universal Motors

A Universal Motor is a series-wound commutator machine designed to run on either direct current (DC) or single-phase alternating current (AC). When powered by AC, current reverses direction simultaneously in both the field coils and the armature, preserving a unidirectional Lorentz force vector ($T \propto I^2$).

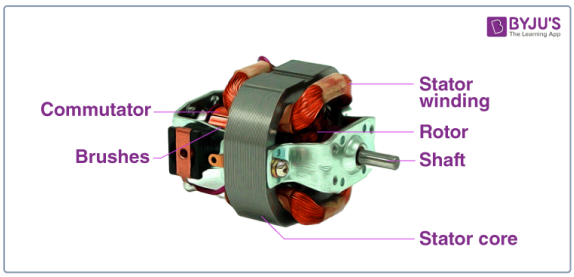


Fig. 11. Universal AC/DC motor showing stator poles, and commutator etc.

These motors deliver exceptionally high operating speeds (often exceeding 10,000–25,000 RPM) and high power-to-weight ratios at low manufacturing cost. However, they share the physical drawbacks of brushed systems, including brush wear, electrical arcing, and acoustic noise. They are common in household blenders, food processors, upright vacuum cleaners, and corded power drills.

B. Linear Motors

Linear motors can be visualized as rotary machines unrolled flat, converting electrical energy directly into linear mechanical translation without intermediate ballscrews, timing belts, or rack-and-pinion linkages. Available in Linear Induction (LIM) and Linear Synchronous (LSM) configurations, they eliminate mechanical backlash and transmission friction. Key applications include magnetic levitation trains (Shanghai Maglev), high-speed gantry stages, semiconductor photolithography tables, and rollercoaster launch mechanisms.

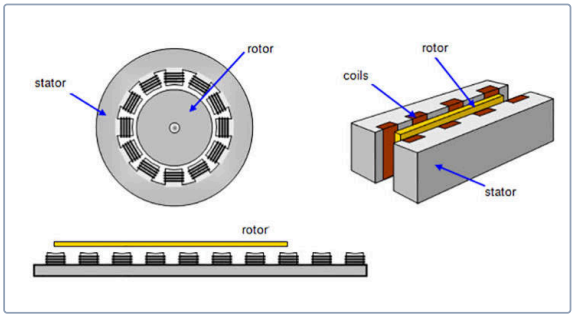


Fig. 12. Linear motors visualized as rotary machines unrolled flat

TABLE I. COMPARISON OF MAJOR ELECTRIC MOTOR TOPOLOGIES

Motor Type	Supply	Efficiency	Control Overhead	Key Trade-Off / Limitation
Brushed DC	DC	70–85%	Lowest (PWM/H-Bridge)	Brush friction, arcing, EMI
BLDC	Switched DC	85–92%	Moderate (Inverter/ESC)	Trapezoidal torque ripple, inverter cost
PMSM	3-Phase AC	92–97%	High (DSP / FOC)	Rare-earth cost, risk of demagnetization
Stepper	Pulsed DC	60–75%	Low-Med (Step/Dir Driver)	Torque drop at high speeds, open-loop step loss
Standard Servo	DC + PWM	70–85%	Low (Analog/Digital PID)	Gear train backlash, potentiometer wear
Smart Servo	DC + Serial	75–90%	Low-Med (Packet Protocol)	Gear backlash, serial bus latency
AC Induction	1/3-Phase AC	85–95%	Low to Moderate (VFD)	Rotor slip losses, high starting inrush current
Synchronous	3-Phase AC + DC	90–97%	High (DC Excitation/Sync)	Requires starting mechanism, dual excitation supplies
SynRM / SRM	Inverter/Pulsed	88–95%	High (Reluctance Profiling)	Acoustic noise, non-linear torque ripple
Universal	AC / DC	65–75%	Lowest (Phase Angle/Triac)	Loud acoustic noise, rapid brush wear, EMI

XII. CONCLUSION

Electric motor topologies balance trade-offs among mechanical complexity, magnetic materials, and power electronics. While brushed DC, universal, and induction machines remain common in cost-sensitive applications, high-performance systems increasingly rely on electronically commutated PMSM, BLDC, and Reluctance motors. Field-Oriented Control and wide-bandgap semiconductors (SiC and GaN) continue to expand the operational envelopes of high-efficiency electric drives across robotics, aerospace, and electric transportation.

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